University of Bologna
Master's Degree in Computer Science and Engineering

Lab-activity-06

Aggregation Behaviour in ARGoS

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0.1 Objective

The objective of this activity is to implement and test a simple aggregation behaviour for a swarm of foot-bots in ARGoS. The goal is to make the robots form clusters by using only local information, without any central controller or global knowledge of the environment.
Each robot runs the same Lua controller and can be in one of two states: `MOVING` or `STOPPED`. The decision to stop or restart depends on the number of stopped neighbours detected through the Range-and-Bearing sensor. In this way, the activity shows how a collective behaviour can emerge from simple probabilistic rules executed independently by each robot.

0.2 Behavior Description

At the beginning of the experiment, every robot is initialized in the `MOVING` state. The robot also sends the value `0` through the Range-and-Bearing actuator, meaning that it is currently moving, and its LEDs are set to green.
During each control step, the robot counts how many neighbouring robots are stopped. This is done by reading the Range-and-Bearing messages and considering only robots within the maximum range `MAXRANGE`. A neighbour is counted only if its transmitted value is equal to `1`, which means that the neighbour is stopped.
If the robot is moving, it computes the probability of stopping as:

$$P_s = \min(PS\_MAX, S + \alpha N)$$

where `N` is the number of stopped neighbours. Therefore, the more stopped robots are nearby, the higher the probability that the current robot will stop too. When the robot stops, it sets its wheels to zero velocity, changes its LED colour to red, and transmits the value `1`.
If the robot is stopped, it computes the probability of starting again as:

$$P_w = \max(PW\_MIN, W - \beta N)$$

In this case, the presence of many stopped neighbours reduces the probability of leaving the cluster. This makes groups more stable once they are formed, while still allowing a small probability of movement because of the lower bound `PW_MIN`.
When a robot is in the `MOVING` state and does not stop, it performs a random walk with obstacle avoidance. The front proximity sensors are divided into left and right groups. If an obstacle is detected, the wheel speeds are adjusted so that the robot turns away from the more occupied side. If no obstacle is detected, the robot usually moves forward, but with a small probability it performs a random turn. This helps the swarm explore the arena and increases the chance of finding other robots.

0.3 Conclusion

The controller works as expected and allows the robots to create groups in the arena. The result, however, depends quite a lot on the chosen parameters. If the probability of stopping grows too slowly, robots continue to move for a long time and stable groups are harder to obtain. If it grows too quickly, robots may stop too early and create many small groups instead of one clearer aggregation.

This is also one of the main aspects of collective choice. The advantage is that the swarm can reach a common behaviour without a central controller, using only local sensing and communication. The disadvantage is that the final result is not always perfectly predictable, because it depends on random movement, local density, and the values of the probabilities.

To bias the aggregation toward a specific area of the arena, the controller could add an extra stopping probability when the robot detects that area, for example by using ground sensors on a black spot. In the same way, the walking probability could be reduced in that area, so robots would be more likely to remain there once they arrive.

Overall, the experiment shows the main trade-off between distributed and centralised solutions. A distributed solution is more scalable and does not fail because of a single central unit, but it is harder to tune and control precisely. A centralised solution would be easier to coordinate, but it would be less robust and less suitable for a swarm where each robot should act using only local information.